

Manage Data in **Azure Storage**

Blob access tiers, replication strategies, lifecycle policies, object replication, import/export jobs, and the tools that move data in and out — everything you need to manage what lives inside a storage account.

Why data management has its own topic

A storage account is not a flat bucket — it has tiers that trade cost against retrieval speed, replication options that control durability and geography, and lifecycle rules that automate transitions and deletions so you're not paying Hot prices for data nobody reads. Understanding these is how you architect low-cost, durable, compliant storage.

Blob access tiers

HOT	COOL	COLD	ARCHIVE
Frequently accessed	Infrequently accessed	Rarely accessed	Long-term offline
Highest storage cost, lowest read cost. Default for new blobs.	Lower storage cost, higher read cost. Min 30-day retention penalty if deleted early.	Even lower storage cost. Min 90-day retention penalty.	Lowest storage cost. Blob is offline — must rehydrate (hours) before reading. Min 180-day retention penalty.

Cost rule: storage cost goes Hot → Cool → Cold → Archive ↓ | access/retrieval cost goes Hot → Cool → Cold → Archive ↑

Replication options

LRS

Locally Redundant

3 copies in one datacenter.
Lowest cost, no zone or region protection.

ZRS

Zone Redundant

3 copies across 3 availability zones in one region. Survives a datacenter outage.

GRS

Geo-Redundant

LRS in primary + async copy to a secondary region. Failover only on Microsoft-initiated disaster declaration.

RA-GRS

Read-Access Geo-Redundant

GRS but secondary is readable at all times — useful for read-heavy DR scenarios.

GZRS

Geo-Zone Redundant

ZRS in primary + async copy to secondary region. Best durability.

RA-GZRS

Read-Access GZRS

GZRS but secondary is also readable. Highest availability.

Eight terms you need cold

Blob Container

A flat namespace within a storage account that holds blobs — similar to a directory but without true hierarchy (paths are just part of the blob name).

Tip: Access level can be set at the container: Private, Blob (anonymous reads), or Container (anonymous reads + list).

Lifecycle Management Policy

A set of rules that automatically move blobs between tiers or delete them based on age or last-modified date.

Tip: Rules apply at the account or container level and run on a daily scan — there's no instant trigger.

Rehydration

Moving a blob out of Archive tier back to Hot or Cool so it can be read. Standard rehydration takes up to 15 hours; High Priority takes under 1 hour (higher cost).

Tip: Archive blobs cannot be read directly — any read attempt fails until rehydration is complete.

Object Replication

Async copy of block blobs from a source container to a destination container, even across regions or subscriptions.

Tip: Requires versioning enabled on both accounts. Does not replicate snapshots or blob metadata changes made after the initial write.

Blob Versioning

Automatically keeps previous versions of a blob when it's overwritten or deleted — each version is addressable by its version ID.

Tip: Required for Object Replication and Change Feed. Versioning adds storage cost.

AzCopy

A command-line utility for high-throughput data movement to and from Azure Storage — blobs, files, and S3 compatible sources.

Tip: Supports resumable transfers and can authenticate with Azure AD or SAS tokens.

Storage Explorer

A cross-platform GUI app for browsing storage accounts, uploading and downloading blobs, and managing containers, queues, and tables.

Tip: Also available as a blade inside the Azure Portal for quick access.

Import/Export Job

A service for shipping physical hard drives to/from Azure datacenters to move very large datasets (petabytes) when network transfer would take too long.

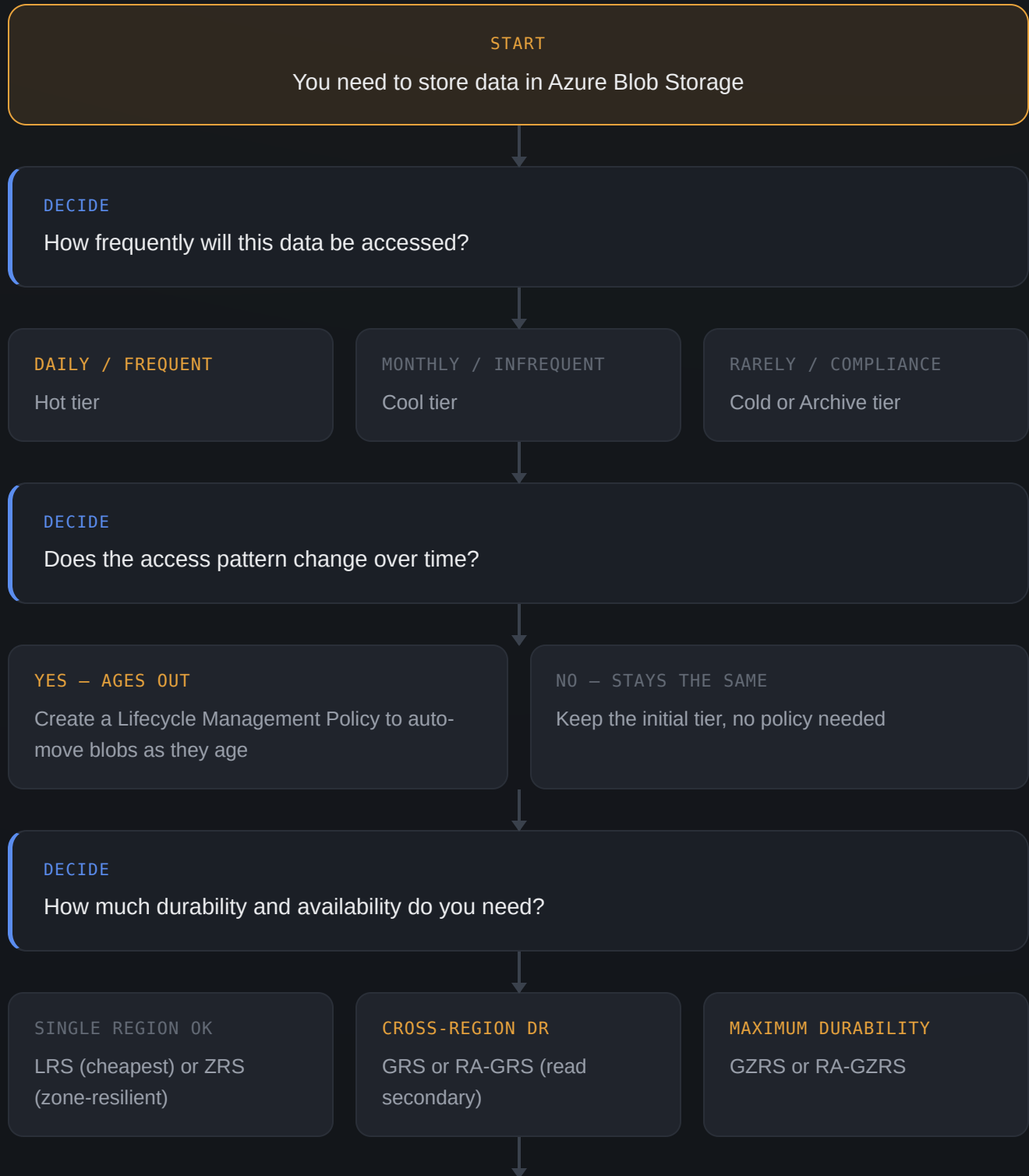
Tip: Use when data size × transfer time exceeds what's practical over the wire — typically multi-TB migrations.

EXAM GOTCHA

The **early deletion penalty** doesn't block the tier change — Azure just **bills you as if the blob had stayed** for the minimum duration (30 days Cool, 90 days Cold, 180 days Archive). The tier change still succeeds.

Managing data lifecycle, tier, and movement

Use this flow when deciding how to store, protect, and eventually expire data in Azure Storage.



DECIDE

Do you need to copy blobs to another account or region?

ONGOING ASYNC COPY

Object Replication (enable versioning on both accounts)

ONE-TIME BULK MOVE

AzCopy or Storage Explorer

MASSIVE DATASET OFFLINE

Azure Import/Export job

DECIDE

Do you need previous versions of blobs if they're overwritten?

YES

Enable Blob Versioning on the account

NO

Skip

DONE

Data stored at the right tier, replicated appropriately, and lifecycle-managed

Tools for managing storage data

Tier changes, lifecycle policies, replication rules, and bulk data movement can all be driven from any of these — pick based on scope and whether it's repeatable.

Azure Portal

BEST FOR: visual lifecycle policy builder, tier changes

The lifecycle management blade has a visual rule builder — the fastest way to understand policy structure before scripting it. Storage Explorer is also embeddable in the portal.

```
Storage Account → Data management
  → Lifecycle management →
Redundancy
  → Object replication
```

AzCopy

BEST FOR: bulk transfers, migrations

The fastest tool for moving large volumes of blobs between accounts, regions, or from S3. Resumable, scriptable, and authenticates via Azure AD or SAS.

```
azcopy copy \

'https://src.blob.core.windows.net/
container?sas' \

'https://dst.blob.core.windows.net/
container?sas' \
  --recursive

azcopy sync \

'https://src.blob.core.windows.net/
container?sas' \

'https://dst.blob.core.windows.net/
container?sas'
```

>_ Azure CLI

BEST FOR: scripting lifecycle policies, tier changes

Change a blob's tier, upload/download files, and manage lifecycle policies in commands that drop into a pipeline script.

```
az storage blob set-tier \  
  --account-name "myaccount" \  
  --container-name "mycontainer" \  
  --name "myblob.txt" --tier Cool  
  
az storage account management-  
policy create \  
  --account-name "myaccount" \  
  --resource-group "rg" \  
  --policy @policy.json
```

>_ Azure PowerShell

BEST FOR: Windows / enterprise automation

Full parity with CLI for tier management, lifecycle policies, and replication setup — fits naturally in Windows-centric pipelines.

```
$blob = Get-AzStorageBlob `   
  -Container "mycontainer" -Blob   
  "myblob.txt" `   
  -Context $ctx  
$blob.ICloudBlob.SetStandardBlobTier("Cool")  
  
Set-  
AzStorageAccountManagementPolicy `   
  -ResourceGroupName "rg" `   
  -StorageAccountName "myaccount" `   
  -Policy $policy
```

⟨⟩ Bicep / ARM

BEST FOR: IaC – lifecycle policies as code

Deploy lifecycle management policies and replication rules alongside the storage account — everything version-controlled and repeatable.

```
resource policy
'Microsoft.Storage/storageAccounts/managementPolicies@2023-01-01' = {
  name: 'default'
  parent: storageAccount
  properties: {
    policy: {
      rules: [{
        name: 'move-to-cool'
        type: 'Lifecycle'
        definition: {
          actions: {
            baseBlob: {
              tierToCool: {
                daysAfterModificationGreaterThan:
                30 }
            }
          }
          filters: { blobTypes:
            ['blockBlob'] }
        }
      }]
    }
  }
}
```


TOOL	BEST FOR	HANDLES BULK TRANSFER?
Portal	Visual policy builder, one-off tier changes	Limited
AzCopy	High-throughput blob movement	Yes — purpose-built
CLI	Scripting, pipeline automation	Small scale
PowerShell	Windows enterprise automation	Small scale
Bicep / ARM	IaC lifecycle + replication policies	No

Principles worth internalizing

Tier intelligently, replicate to match your SLA, and automate everything you can with lifecycle policies.

REPLICATION	PROTECTS AGAINST	READ SECONDARY?	TYPICAL USE CASE
LRS	Hardware failure in one rack	No	Dev/test, easily reproducible data
ZRS	Datacenter / zone outage	No	High-availability within a region
GRS	Full region outage	No (failover only)	Business-critical production data
RA-GRS	Full region outage	Yes — always readable	Read-heavy DR workloads
GZRS	Zone outage + region outage	No (failover only)	Mission-critical, highest durability

LIFECYCLE POLICY TIPS

Rules run on a **daily scan** — there's no real-time trigger. Use `daysAfterModificationGreaterThan` for transition actions and `daysAfterCreationGreaterThan` for delete rules. Apply filters by blob type, prefix, or index tags to keep rules narrow. **Archive blobs must rehydrate before any read or tier change back to Hot/Cool can complete.**

Six mistakes that show up on the exam

Reading an Archive blob directly

Archive blobs are offline. Any read attempt fails until rehydration (up to 15 hours) is complete.

Expecting lifecycle changes to be instant

The lifecycle engine runs a daily scan — tier transitions happen within 24 hours, not immediately.

Ignoring early-deletion penalties

Moving a blob out of Cool (30 days), Cold (90 days), or Archive (180 days) before minimum age incurs a prorated penalty.

Setting up Object Replication without versioning

Object Replication requires blob versioning enabled on both source and destination accounts — it silently won't work without it.

Choosing GRS expecting a readable secondary

GRS secondary is only readable during a Microsoft-initiated failover — for always-readable DR, use RA-GRS.

Using AzCopy copy vs. sync incorrectly

`copy` always transfers all objects; `sync` only transfers changed or missing objects — use `sync` for ongoing mirror jobs.